- A therapeutic vaccine that increases the immune response against the HIV virus.
- A vaccine that protects infants against meningococcal disease, a leading cause of meningitis.
- An immunotherapeutic vaccine for the treatment of Alzheimer's disease.
- A recombinant vaccine to prevent malaria.

Evolving science has increasingly enabled researchers to explore both promising therapeutic vaccines and new preventative agents for infectious diseases. Although the development process is extremely complex, advances in other scientific fields, such as genomics, are being leveraged in the development of new vaccines.

"Vaccines have been a major contributor in saving countless lives around the world," said Castellani. "Vaccinations contribute to the public health at large, and they make good economic sense. The many exciting candidates in the pipeline offer great hope for a healthier, more productive future."

1.3.1 Monoclonal Antibodies

mAb or moAb are identical immunoglobulins, generated from a single B-cell clone. These antibodies recognize unique epitopes, or binding sites, on a single antigen. Derivation from a single B-cell clones and subsequent targeting of a single epitope is what differentiates monoclonal antibodies from polyclonal antibodies

The traditional monoclonal antibody (mAb) production process usually starts with generation of mAb-producing cells (i.e. hybridomas) by fusing myeloma cells with desired antibody-producing splenocytes (Example: B cells). These B cells are typically sourced from animals, usually mice. After cell fusion, large numbers of clones are screened and selected on the basis of antigen specificity and immunoglobulin class (Figure 1.3).

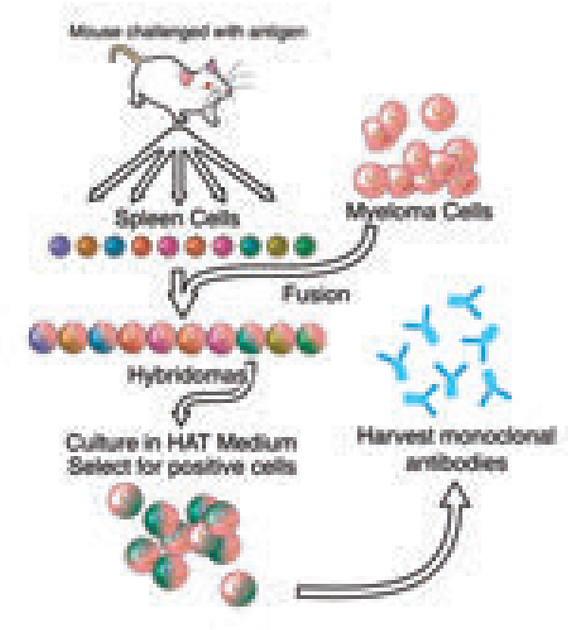


Figure 1.3: Monoclonal antibody

1.3.2 Stem Cell & Therapy

Stem cells are biological cells that can differentiate into other types of cells & they are found in multicellular organism. Stem cells are a class of undifferentiated cells that are able to differentiate into specialized cell types. Commonly, stem cells come from two main sources:

- Embryos formed during the blastocyst phase of embryological development (embryonic stem cells) and
- Adult tissue (adult stem cells).

Both types are generally characterized by their potency, or potential to

TN_GOVT_XII_Micro_Biology_CH01.indd 5 27-02-2019 15:14:48